Chapter 03
Numerical Initial Value Problem

Lesson 3

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3.2. Linear Multistep Methods

Runge-Kutta Method

Runge Kutta method (RK4)

Consider the usual initial value problem

$$\frac{dy}{dt} = f(t, y), \qquad y(0) = y_0.$$

$$\begin{aligned}
\text{Calculate} \quad K_1 &= f(nh, y_n) \\
\text{then} \quad K_2 &= f((n + \tfrac{1}{2})h, y_n + \tfrac{1}{2}hK_1) \\
\text{then} \quad K_3 &= f((n + \tfrac{1}{2})h, y_n + \tfrac{1}{2}hK_2) \\
\text{then} \quad K_4 &= f((n + 1)h, y_n + hK_3) \\
\text{finally} \quad y_{n+1} &= y_n + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4)
\end{aligned}$$

Example 01

Suppose that $y = y(t)$ is the solution to the initial value problem

$$\frac{dy}{dt} = \cos(y) \qquad y(0) = 3$$

Carry out one time step of the Runge-Kutta method RK4 with a step size of $h = 0.1$ so as to obtain an approximation to $y(0.1)$.

Solution

The iteration must be carried out in four stages. We start by calculating

$$K_1 = f(0, y_0) = f(0, 3) = -0.989992$$

a value we now use in finding

$$K_2 = f(\tfrac{1}{2}h, y_0 + \tfrac{1}{2}hK_1) = f(0.05, 2.950500) = -0.981797$$

This value K_2 is now used in our evaluation of

$$K_3 = f(\tfrac{1}{2}h, y_0 + \tfrac{1}{2}hK_2) = f(0.05, 2.950910) = -0.981875$$

which, in turn, is used in

$$K_4 = f(h, y_0 + hK_3) = f(0.1, 2.901812) = -0.971390$$

All four of these values are then used to complete the iteration

$$\begin{aligned} y_1 &= y_0 + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4) \\ &= 3 + \frac{0.1}{6}(-0.989992 + 2 \times -0.981797 + 2 \times -0.981875 - 0.971390) \\ &= 2.901855 \qquad \text{to 6 decimal places.} \end{aligned}$$

Example 02

Suppose that $y = y(t)$ is the solution to the initial value problem

$$\frac{dy}{dt} = y(1-y) \qquad y(0) = 0.7$$

Carry out one time step of the Runge-Kutta method RK4 with a step size of $h = 0.1$ so as to obtain an approximation to $y(0.1)$.

3.3. Predictor- Corrector Methods

- In this final Section on numerical approximations for initial value problems involving ordinary differ- entia' equations we consider predictor-corrector methods.

  These methods are a way of getting around the difficulties inherent in implementing certain implicit numerical schemes.

We have seen that when using an implicit linear multistep method there is an additional difficulty because we cannot, in general, solve simply for the newest approximate y-value y_{n+k}. A general k-step implicit method involves, at the kth time step,

$$\underset{\substack{\uparrow \\ \text{the} \\ \text{unknown}}}{\alpha_k y_{n+k}} + \cdots + \alpha_1 y_1 + \alpha_0 y_0 = h(\underset{\substack{\uparrow \\ y_{n+k} \text{ occurs} \\ \text{here too}}}{\beta_k f_{n+k}} + \cdots + \beta_1 f_1 + \beta_0 f_0)$$

and if f depends on y in a complicated way then it is not obvious how to dig y_{n+k} out of $f_{n+k} = f((n+k)h, y_{n+k})$.

One solution to this problem would be to only ever use **explicit** methods in which $\beta_k = 0$. But this is not a good solution, for **implicit** methods generally have better properties than the explicit ones (for example, the implicit trapezium is second order while the explicit Euler is only first order). Another solution involves a so-called **predictor-corrector** method. This involves

1. The predictor step. We use an explicit method to obtain an approximation y^P_{n+k} to y_{n+k}.

2. The corrector step. We use an implicit method, but with the predicted value y^P_{n+k} on the right-hand side in the evaluation of f_{n+k}. We use f^P_{n+k} to denote this approximate (predicted) value of f_{n+k}.

3. We can then go on to correct again and again. At each step we put the latest approximation to y_{n+k} in the right-hand side of the scheme (*via* f) to generate a new approximation from the left-hand side.

It is sufficient for our purposes to illustrate the idea of a predictor-corrector method using the simplest possible pair of methods. We use Euler's method to predict and the trapezium method to correct

Example 01

Suppose that $y = y(t)$ is the solution to the initial value problem

$$\frac{dy}{dt} = t + y, \qquad y(0) = 3$$

Use Euler's method and the trapezium method as a predictor-corrector pair (with one correction at each time step). Take the time step to be $h = 0.05$ so as to obtain approximations to $y(0.05)$ and $y(0.1)$.

Solution

Euler's method, $y_{n+1} = y_n + hf_n$, is the explicit method so we use that to **predict**. For the first time step we require $f_0 = f(0, y_0) = f(0, 3) = 3$ and therefore

$$y_1^P = y_0 + hf_0 = 3 + 0.05 \times 3 = 3.15$$

We now use this predicted value of y_1 to obtain a "predicted" value for f_1 which we can use in the implicit trapezium method. We find $f_1^P = f(h, y_1^P) = f(0.05, 3.15) = 3.2$. We now **correct** using the trapezium method in the form

$$y_1 = y_0 + \frac{h}{2}\left(f_0 + f_1^P\right) = 3 + \frac{1}{2}(0.05)(3 + 3.2) = 3.155$$

This completes prediction and one correction for the first time step.
For the second time step we require $f_1 = f(h, y_1) = f(0.05, 3.155) = 3.205$ and therefore

$$y_2^P = y_1 + hf_1 = 3.155 + 0.05 \times 3.205 = 3.31525$$

which is the predicted value for y_2. We now correct it with

$$y_2 = y_1 + \frac{h}{2}\left(f_1 + f_2^P\right) = 3.155 + \frac{1}{2}(0.05)(3.205 + 3.41525) = 3.320506$$

We conclude that

$$\begin{aligned} y(0.05) &\approx 3.155 \\ y(0.1) &\approx 3.320506 \end{aligned}$$

Example 02

Suppose that $y = y(t)$ is the solution to the initial value problem

$$\frac{dy}{dt} = -\tan(y) \qquad y(0) = 1$$

Use Euler's method and the trapezium method as a predictor-corrector pair (with one correction at each time step). Take the time step to be $h = 0.2$ so as to obtain approximations to $y(0.2)$ and $y(0.4)$.

Solution

Euler's method, $y_{n+1} = y_n + hf_n$, is the explicit method so we use that to predict. For the first time step we require $f_0 = f(0, y_0) = f(0, 1) = -1.55741$ and therefore

$$y_1^P = y_0 + hf_0 = 1 + 0.2 \times -1.55741 = 0.688518$$

We now use this predicted value to obtain a "predicted" value for f_1 which we can use in the implicit trapezium method. We find $f_1^P = f(h, y_1^P) = f(0.2, 0.688518) = -0.82285$. We now correct using the trapezium method in the form

$$y_1 = y_0 + \frac{h}{2}\left(f_0 + f_1^P\right) = 1 + \frac{1}{2}(0.2)(-1.55741 - 0.822848) = 0.761974$$

This completes prediction and one correction for the first time step.

For the second time step we require $f_1 = f(h, y_1) = f(0.2, 0.761974) = -0.95422$ and therefore

$$y_2^P = y_1 + hf_1 = 0.76194 + 0.2 \times -0.95422 = 0.571131$$

which is the predicted value for y_2. We now correct it with

$$y_2 = y_1 + \frac{h}{2}\left(f_1 + f_2^P\right) = 0.761974 + \frac{1}{2}(0.2)(-0.95422 - -0.64257) = 0.602296$$

We conclude that

$$\begin{aligned} y(0.2) &\approx 0.761974 \\ y(0.4) &\approx 0.602296 \end{aligned}$$

Example 03

Suppose that $y = y(t)$ is the solution to the initial value problem

$$\frac{dy}{dt} = \cos(y), \qquad y(0) = 0$$

Use Euler's method and the trapezium method as a predictor-corrector pair (with one correction at each time step). Take the time step to be $h = 0.1$ so as to obtain approximations to $y(0.1)$ and $y(0.2)$.